

The central limit theorem implies that the sum of n independently distributed random variables is approximately normal, regardless of the distributions of the individual variables. The approximation improves as n increases. In many cases the approximation will be good for small n —say, $n < 10$ —whereas in some cases we may require very large n —say, $n > 100$ —for the approximation to be satisfactory. In general, if the x_i are identically distributed, and the distribution of each x_i does not depart radically from the normal, then the central limit theorem works quite well for $n \geq 3$ or 4. These conditions are met frequently in quality-engineering problems.

3.3.2 The Lognormal Distribution

Variables in a system sometimes follow an exponential relationship, say $x = \exp(w)$. If the exponent is a random variable, say w , $x = \exp(w)$ is a random variable and the distribution of x is of interest. An important special case occurs when w has a normal distribution. In that case, the distribution of x is called a **lognormal distribution**. The name follows from the transformation $\ln(x) = w$. That is, the natural logarithm of x is normally distributed.

Probabilities for x are obtained from the transformation to w , but we need to recognize that the range of x is $(0, \infty)$. Suppose that w is normally distributed with mean θ and variance ω^2 ; then the cumulative distribution function for x is

$$\begin{aligned} F(a) &= P[x \leq a] = P[\exp(w) \leq a] = P[w \leq \ln(a)] \\ &= P\left[z \leq \frac{\ln(a) - \theta}{\omega}\right] = \Phi\left[\frac{\ln(a) - \theta}{\omega}\right] \end{aligned}$$

for $x > 0$, where z is a standard normal random variable. Therefore, Appendix Table II can be used to determine the probability. Also, $f(x) = 0$, for $x \leq 0$. The lognormal random variable is always nonnegative.

The lognormal distribution is defined as follows.

Definition

Let w have a normal distribution mean θ and variance ω^2 ; then $x = \exp(w)$ is a **lognormal random variable**, and the lognormal distribution is

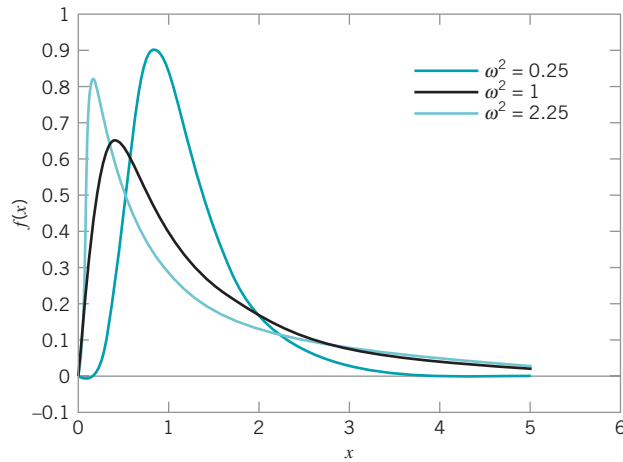
$$f(x) = \frac{1}{x\omega\sqrt{2\pi}} \exp\left[-\frac{(\ln(x) - \theta)^2}{2\omega^2}\right] \quad 0 < x < \infty \quad (3.29)$$

The mean and variance of x are

$$\mu = e^{\theta + \omega^2/2} \quad \text{and} \quad \sigma^2 = e^{2\theta + 2\omega^2} (e^{\omega^2} - 1) \quad (3.30)$$

The parameters of a lognormal distribution are θ and ω^2 , but care is needed to interpret that these are the mean and variance of the normal random variable w . The mean and variance of x are the functions of these parameters shown in equation (3.30). Figure 3.20 illustrates lognormal distributions for selected values of the parameters.

The lifetime of a product that degrades over time is often modeled by a lognormal random variable. For example, this is a common distribution for the lifetime of a semiconductor laser. Other continuous distributions can also be used in this type of application. However, because the lognormal distribution is derived from a simple exponential function of a normal random variable, it is easy to understand and easy to evaluate probabilities.



■ **FIGURE 3.20** Lognormal probability density functions with $\theta = 0$ for selected values of ω^2 .

EXAMPLE 3.10 Medical Laser Lifetime

The lifetime of a medical laser used in ophthalmic surgery has a lognormal distribution with $\theta = 6$ and $\omega = 1.2$ hours. What is the probability the lifetime exceeds 500 hours?

SOLUTION

From the cumulative distribution function for the lognormal random variable

$$\begin{aligned} P(x > 500) &= 1 - P[\exp(w) \leq 500] = 1 - P[w \leq \ln(500)] \\ &= \Phi\left(\frac{\ln(500) - 6}{1.2}\right) = 1 - \Phi(0.1788) \\ &= 1 - 0.5710 = 0.4290 \end{aligned}$$

What lifetime is exceeded by 99% of lasers? Now the question is to determine a such that $P(x > a) = 0.99$. Therefore,

$$\begin{aligned} P(x > a) &= P[\exp(w) > a] = P[w > \ln(a)] \\ &= 1 - \Phi\left(\frac{\ln(a) - 6}{1.2}\right) = 0.99 \end{aligned}$$

From Appendix Table II, $1 - \Phi(a) = 0.99$ when $a = -2.33$. Therefore,

$$\frac{\ln(a) - 6}{1.2} = -2.33 \quad \text{and} \quad a = \exp(3.204) = 24.63 \text{ hours}$$

Determine the mean and standard deviation of lifetime. Now,

$$\begin{aligned} \mu &= e^{\theta + \omega^2/2} = \exp(6 + 0.72) = 828.82 \text{ hours} \\ \sigma^2 &= e^{2\theta + \omega^2} (e^{\omega^2} - 1) = \exp(12 + 1.44) [\exp(1.44) - 1] \\ &= 2,212,419.85 \end{aligned}$$

so the standard deviation of the lifetime is 1487.42 hours. Notice that the standard deviation of the lifetime is large relative to the mean.

3.3.3 The Exponential Distribution

The probability distribution of the exponential random variable is defined as follows.

Definition

The **exponential distribution** is

$$f(x) = \lambda e^{-\lambda x} \quad x \geq 0 \quad (3.31)$$

where $\lambda > 0$ is a constant. The **mean** and **variance** of the exponential distribution are

$$\mu = \frac{1}{\lambda} \quad (3.32)$$

and

$$\sigma^2 = \frac{1}{\lambda^2} \quad (3.33)$$

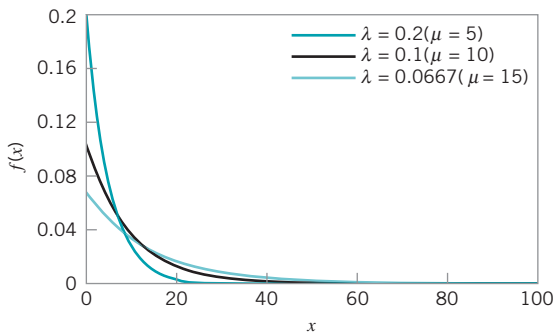
respectively.

Several exponential distributions are shown in Fig. 3.21.

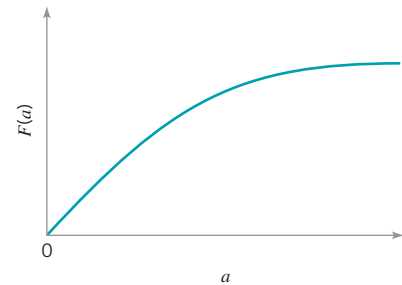
The cumulative exponential distribution is

$$\begin{aligned} F(a) &= P\{x \leq a\} \\ &= \int_0^a \lambda e^{-\lambda t} dt \\ &= 1 - e^{-\lambda a} \quad a \geq 0 \end{aligned} \quad (3.34)$$

Figure 3.22 illustrates the exponential cumulative distribution function.



■ **FIGURE 3.21** Exponential distributions for selected values of λ .



■ **FIGURE 3.22** The cumulative exponential distribution function.

The exponential distribution is widely used in the field of **reliability engineering** as a model of the time to failure of a component or system. In these applications, the parameter λ is called the **failure rate** of the system, and the mean of the distribution $1/\lambda$ is called the **mean time to failure**.² For example, suppose that an electronic component in an airborne radar system has a useful life described by an exponential distribution with failure rate $10^{-4}/\text{h}$; that is, $\lambda = 10^{-4}$. The mean time to failure for this component is $1/\lambda = 10^4 = 10,000$ h. If we wanted to determine the probability that this component would fail before its expected life, we would evaluate

$$P\left\{x \leq \frac{1}{\lambda}\right\} = \int_0^{1/\lambda} \lambda e^{-\lambda t} dt = 1 - e^{-1} = 0.63212$$

This result holds regardless of the value of λ ; that is, the probability that a value of an exponential random variable will be less than its mean is 0.63212. This happens, of course, because the distribution is not symmetric.

There is an important relationship between the exponential and Poisson distributions. If we consider the Poisson distribution as a model of the number of occurrences of some event in the interval $(0, t]$, then from equation 3.15 we have

$$p(x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

Now $x = 0$ implies that there are no occurrences of the event in $(0, t]$, and $P\{x = 0\} = p(0) = e^{-\lambda t}$. We may think of $p(0)$ as the probability that the interval to the first occurrence is greater than t , or

$$P\{y > t\} = p(0) = e^{-\lambda t}$$

where y is the random variable denoting the interval to the first occurrence. Since

$$F(t) = P\{y \leq t\} = 1 - e^{-\lambda t}$$

and using the fact that $f(y) = dF(y)/dy$, we have

$$f(y) = \lambda e^{-\lambda y} \tag{3.35}$$

as the distribution of the interval to the first occurrence. We recognize equation (3.35) as an exponential distribution with parameter λ . Therefore, we see that if the number of occurrences of an event has a Poisson distribution with parameter λ , then the distribution of the interval *between* occurrences is exponential with parameter λ .

3.3.4 The Gamma Distribution

The probability distribution of the gamma random variable is defined as follows.

²See the supplemental text material for more information.

Definition

The **gamma distribution** is

$$f(x) = \frac{\lambda}{\Gamma(r)} (\lambda x)^{r-1} e^{-\lambda x} \quad x \geq 0 \quad (3.36)$$

with **shape parameter** $r > 0$ and **scale parameter** $\lambda > 0$. The **mean** and **variance** of the gamma distribution are

$$\mu = \frac{r}{\lambda} \quad (3.37)$$

and

$$\sigma^2 = \frac{r}{\lambda^2} \quad (3.38)$$

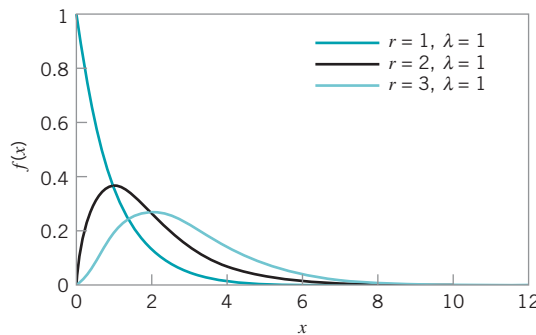
respectively.³

Several gamma distributions are shown in Fig. 3.23. Note that if $r = 1$, the gamma distribution reduces to the exponential distribution with parameter λ (Section 3.3.3). The gamma distribution can assume many different shapes, depending on the values chosen for r and λ . This makes it useful as a model for a wide variety of continuous random variables.

If the parameter r is an integer, then the gamma distribution is the sum of r independently and identically distributed exponential distributions, each with parameter λ . That is, if $x_1, x_2, \dots, x_r$ are exponential with parameter λ and independent, then

$$y = x_1 + x_2 + \dots + x_r$$

is distributed as gamma with parameters r and λ . There are a number of important applications of this result.

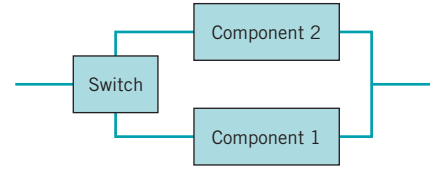


■ **FIGURE 3.23** Gamma distributions for selected values of r and $\lambda = 1$.

³ $\Gamma(r)$ in the denominator of equation 3.36 is the gamma function, defined as $\Gamma(r) = \int_0^{\infty} x^{r-1} e^{-x} dx$, $r > 0$. If r is a positive integer, then $\Gamma(r) = (r-1)!$

EXAMPLE 3.11 A Standby Redundant System

Consider the system shown in Fig. 3.24. This is called a **standby redundant system**, because while component 1 is on, component 2 is off, and when component 1 fails, the switch automatically turns component 2 on. If each component has a life described by an exponential distribution with $\lambda = 10^{-4}/\text{h}$, say, then the system life is gamma distributed with parameters $r = 2$ and $\lambda = 10^{-4}$. Thus, the mean time to failure is $\mu = r/\lambda = 2/10^{-4} = 2 \times 10^4 \text{ h}$.



■ **FIGURE 3.24** The standby redundant system for Example 3.11.

The cumulative gamma distribution is

$$F(a) = 1 - \int_a^\infty \frac{\lambda}{\Gamma(r)} (\lambda t)^{r-1} e^{-\lambda t} dt \quad (3.39)$$

If r is an integer, then equation 3.39 becomes

$$F(a) = 1 - \sum_{k=0}^{r-1} e^{-\lambda a} \frac{(\lambda a)^k}{k!} \quad (3.40)$$

Consequently, the cumulative gamma distribution can be evaluated as the sum of r Poisson terms with parameter λa . This result is not too surprising, if we consider the Poisson distribution as a model of the number of occurrences of an event in a fixed interval, and the gamma distribution as the model of the portion of the interval required to obtain a specific number of occurrences.

3.3.5 The Weibull Distribution

The Weibull distribution is defined as follows.

Definition

The **Weibull distribution** is

$$f(x) = \frac{\beta}{\theta} \left(\frac{x}{\theta} \right)^{\beta-1} \exp \left[- \left(\frac{x}{\theta} \right)^\beta \right] \quad x \geq 0 \quad (3.41)$$

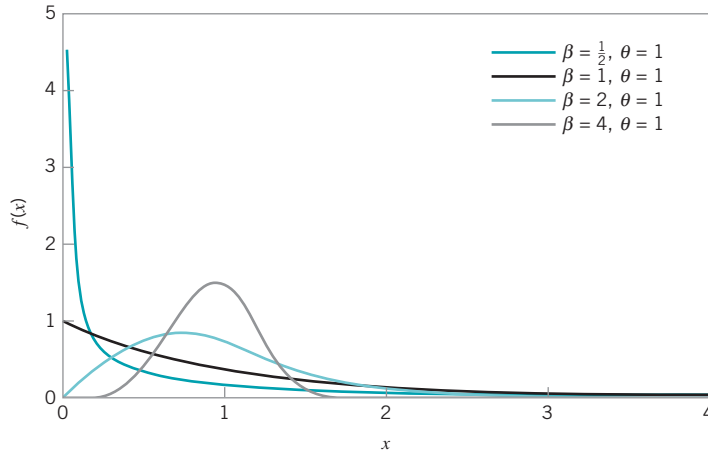
where $\theta > 0$ is the **scale parameter**, and $\beta > 0$ is the **shape parameter**. The **mean** and **variance** of the Weibull distribution are

$$\mu = \theta \Gamma \left(1 + \frac{1}{\beta} \right) \quad (3.42)$$

and

$$\sigma^2 = \theta^2 \left[\Gamma \left(1 + \frac{2}{\beta} \right) - \left\{ \Gamma \left(1 + \frac{1}{\beta} \right) \right\}^2 \right] \quad (3.43)$$

respectively.



■ **FIGURE 3.25** Weibull distributions for selected values of the shape parameter β and scale parameter $\theta = 1$.

The Weibull distribution is very flexible, and by appropriate selection of the parameters θ and β , the distribution can assume a wide variety of shapes. Several Weibull distributions are shown in Fig. 3.25 for $\theta = 1$ and $\beta = 1/2, 1, 2$, and 4 . Note that when $\beta = 1$, the Weibull distribution reduces to the exponential distribution with mean $1/\theta$. The cumulative Weibull distribution is

$$F(a) = 1 - \exp \left[- \left(\frac{a}{\theta} \right)^\beta \right] \quad (3.44)$$

The Weibull distribution has been used extensively in reliability engineering as a model of time to failure for electrical and mechanical components and systems. Examples of situations in which the Weibull has been used include electronic devices such as memory elements, mechanical components such as bearings, and structural elements in aircraft and automobiles.⁴

EXAMPLE 3.12 Time to Failure for Electronic Components

The time to failure for an electronic component used in a flat panel display unit is satisfactorily modeled by a Weibull distribution with $\beta = \frac{1}{2}$ and $\theta = 5000$. Find the mean time to failure

and the fraction of components that are expected to survive beyond 20,000 hours.

SOLUTION

The mean time to failure is

$$\begin{aligned} \mu &= \theta \Gamma \left(1 + \frac{1}{\beta} \right) = 5000 \Gamma \left(1 + \frac{1}{\frac{1}{2}} \right) \\ &= 5000 \Gamma(3) = 10,000 \text{ hours} \end{aligned}$$

The fraction of components expected to survive $a = 20,000$ hours is

$$1 - F(a) = \exp \left[- \left(\frac{a}{\theta} \right)^\beta \right]$$

or

$$\begin{aligned} 1 - F(20,000) &= \exp \left[- \left(\frac{20,000}{5,000} \right)^{\frac{1}{2}} \right] \\ &= e^{-2} \\ &= 0.1353 \end{aligned}$$

That is, all but about 13.53% of the subassemblies will fail by 20,000 hours.

⁴See the supplemental text material for more information.